Minh-Quan Duong

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Education

University of Science, VNU-HCM 2024 - Present
Bachelor of Science (Honors Program) in Information Technology
GPA: **9.52/10.00**

Honors & Awards

- Faculty of Information Technology Dean's List– **Top 1**, University of Science, VNU-HCM. 2024-2025
- **Academic Achievement Scholarship**, University of Science, VNU-HCM 2024-2025
- **Top 2**, Final Project DANIT DS 5+6 competition; public leaderboard score: **0.80182**. Jul 2026
- **15th place**, AI City Challenge 2026 Track 2 public leaderboard, Team Adrenaline. Aug 2026
- **Award Winner** – AI Challenge 2026 | Team AGI Oct 2026

Publications

- **M.-Q. Duong**, T.-N. Tran, K.-O. Nguyen-Ngoc, et al. *Representation-Aware Knowledge Distillation for Efficient Malware Family Classification.*To appear, MAPR 2026.

Research Experience

Representation-Aware Knowledge Distillation for Efficient Malware Family Classification
Team lead & First Author | University of Science, VNU-HCM *2026*
Python, PyTorch, Malware Visualization, Transfer Learning, Knowledge Distillation

- Investigated malware family classification from visual representations using transfer learning and knowledge distillation.
- Developed and evaluated three malware representations; documented the approach in a first-authored paper.

VidQuest: Interactive Multimodal Video Retrieval with Context-Aware Temporal Alignment
Co-first Author | Under review, MMM 2027 *2026*
SigLIP2, BGE, FAISS, BM25, OCR, ASR, Temporal Alignment

- Developed interactive retrieval for ordered, same-video events using visual, caption, OCR, and transcript evidence.
- Combined context-guided candidate recovery, temporal alignment, and event-level feedback; reached 82.50% Sequence Success@1 in an internal 40-query validation.

Selected Projects

AI City Challenge 2026 - Traffic Video Understanding Aug 2026
Qwen2.5-VL-7B-Instruct, QLoRA, Visual Question Answering, Video Captioning

- Contributed to a multi-source vision-language framework for traffic video captioning and question answering in a simulation-to-reality setting.
- Standardized preprocessing and coordinated data flow across inference modules; implemented Agent 01 and built data, prior, and rule-based fallback components for Agent 08.

Multi-Task Fashion Review Understanding Jul 2026
DeBERTa-v3, PyTorch, Ordinal Regression, Rank Ensemble

- Contributed to joint prediction of product ratings (1-5) and recommendation labels using DeBERTa-v3, task-adaptive pretraining, ordinal regression, R-Drop, and FGM.
- Implemented nested out-of-fold calibration, rank blending, and distribution-preserving ensembles; audited submissions and managed artifacts to prevent data leakage.

Real-Time Remote Computer Control System Dec 2025
C++, WebSocket, Boost.Asio, Boost.Beast, OpenCV, React

- Built a remote computer management system with asynchronous networking, persistent WebSocket connections, and real-time screen capture, encoding, and streaming.
- Implemented process, application, and file management, clipboard access, and LAN discovery; integrated a web interface for authentication, session management, monitoring, and remote control.

Technical Skills

Programming: Python, C/C++, SQL, JavaScript / TypeScript, HTML, CSS
Libraries: NumPy, Pandas, SciPy, OpenCV, Matplotlib, Pillow, Sentence-Transformers, timm, FAISS
AI Methods: Machine Learning, Deep Learning, Computer Vision, NLP, Vision-Language Models, Knowledge Distillation
Systems & APIs: WebSocket, TCP/IP, Boost.Asio, Boost.Beast, WinAPI, REST APIs, FastAPI, Docker
Research Interests: Multimodal, Computer Vision.

Certifications & Languages

Machine Learning Specialization – DeepLearning.AI & Stanford Online, Coursera (Andrew Ng) Aug 2026
Vietnamese (Native) | English (Professional Working Proficiency – IELTS 6.0)